Conveniences and Performance Charts

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The following notes (based on the lecture of 3 January of the year of the lecture) **Recursion Theory and Computation** and **Recursion and Computation** are intended to be a guide to the lecture. They are not to be considered as a substitute for the lecture and are not to be quoted. Nevertheless, they are intended to be complete.

The lecture is divided into three parts:

- The notion of the computable function, where the argument is assumed to be a sequence of finite or infinite binary strings, and where, further, the function is also assumed to be a function whose values are finite or infinite binary strings. This notion is due to Turing, and I shall call it the Turing notion, or, for short, the Turing computable. The Turing notion is the most convenient for the theory of computation, but it is not the most natural for the theory of recursive functions, and, probably, it is not the most natural for any application of the theory.
- The notion of the recursive function, where the argument is assumed to be a sequence of finite or infinite binary strings, and where, further, the function is also assumed to be a function whose values are finite or infinite binary strings. This notion is due to Kleene, and I shall call it the Kleene notion, or, for short, the Kleene recursive. The Kleene notion is the most natural for the theory of recursive functions, and, probably, it is the most natural for any application of the theory.
- The notion of the recursive function, where the argument is assumed to be a finite or infinite binary string, and where, further, the function is also assumed to be a function whose values are finite or infinite binary strings. This notion is due to Scott, and I shall call it the Scott notion, or, for short, the Scott recursive. The Scott notion is the most convenient for the theory of computation, and, probably, it is the most natural for any application of the theory.

The lecture is a lecture. Charts are not intended to be read, but are intended to be illustrated by a lecture. It is therefore desirable that the Scott chart accompanied the Kleene chart.

In addition, the lecture illustrates a certain dependence of computability on the number of functions. Thus, the notion that is the Scott and Kleene, plus a finite number of functions, is equivalent to the Turing notion. The notion that is the Scott and Kleene, plus the functions of the Kleene chart, is equivalent to the Scott notion. Thus, the Kleene chart is a chart of the Scott notion, plus the functions of the Kleene chart.

- Consider $\{\alpha : \alpha \in 2^\omega\}$ with an open set $\{\alpha \in 2^\omega : \alpha \text{ contains the prefix } \sigma\}$ for each $\sigma \in 2^{<\omega}$.
- Let $\{\sigma_n : n \in \omega\}$ be a recursive enumeration of $2^{<\omega}$.
- Let $\{\alpha_n : n \in \omega\}$ be a recursive enumeration of 2^ω.

Charts with a number in parentheses are supposed to be a Kleene chart, plus, or, and, or, and, or, and